

2022 Ph H2 Q5

Section: Particles and Waves
Topic: Doppler Effect, Redshift, Gravitation

Summary:

This question explores the Doppler effect in sound and light, redshift analysis in a binary star system, and gravitational force between stars at different distances.

(a)

Explain, in terms of wavefronts, why the frequency is lower as the buzzer moves away from the students.

Answer:

- As the buzzer moves **away** from the students, it emits wavefronts that are **more spread out**, increasing the wavelength.
- This causes the students to receive wavefronts **less frequently**, resulting in a **lower frequency** being heard.
- Conversely, when the buzzer moves **towards** the students, the wavefronts are **compressed**, decreasing the wavelength and increasing the frequency.

A diagram showing:

- wavefronts closer together **in front of** the moving source
- wavefronts farther apart **behind** the source

is often used to support the explanation.

Marks: 2

- 1 mark: description of wavefront spacing
- 1 mark: link to observed frequency change

(b)(i)

Calculate the redshift of star B.

Given:

$$\lambda_{\text{rest}} = 580 \text{ nm}, \quad \lambda_{\text{observed}} = 610 \text{ nm}$$
$$z = \frac{\lambda_{\text{observed}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}} = \frac{610 - 580}{580} = \frac{30}{580} \approx 0.0517$$

Answer: 0.052 (2 significant figures)

Mark: 1

(b)(ii)

Determine the approximate distance from Earth to the binary star system.

Use:

$$z = \frac{v}{c} \Rightarrow v = zc = 0.0517 \times 3.0 \times 10^8 \approx 1.55 \times 10^7 \text{ m s}^{-1}$$

Use Hubble's Law:

$$v = H_0 d \Rightarrow d = \frac{v}{H_0}$$

Take $H_0 = 2.3 \times 10^{-18} \text{ s}^{-1}$:

$$d = \frac{1.55 \times 10^7}{2.3 \times 10^{-18}} \approx 6.74 \times 10^{24} \text{ m}$$

Answer: $6.7 \times 10^{24} \text{ m}$

Marks: 2

- 1 mark: correct use of $z = v/c$
- 1 mark: correct use of Hubble's Law and final answer

(c)(i)

Calculate the gravitational force between the two stars.

Given:

$$r = 3.44 \times 10^{12} \text{ m}, \quad m_1 = 2.19 \times 10^{30} \text{ kg}, \quad m_2 = 1.80 \times 10^{30} \text{ kg}$$
$$F = G \frac{m_1 m_2}{r^2}, \quad G = 6.67 \times 10^{-11}$$
$$F = \frac{6.67 \times 10^{-11} \times 2.19 \times 10^{30} \times 1.80 \times 10^{30}}{(3.44 \times 10^{12})^2} \approx \frac{2.63 \times 10^{50}}{1.18 \times 10^{25}} \approx 2.23 \times 10^{25} \text{ N}$$

Answer: $2.2 \times 10^{25} \text{ N}$

Marks: 3

- 1 mark: correct substitution into Newton's law
- 1 mark: correct squaring of distance
- 1 mark: correct final value

(c)(ii)

If the distance between the stars is halved, how many times greater is the gravitational force?

Answer:

- Gravitational force is **inversely proportional to the square of the distance**:

$$F \propto \frac{1}{r^2} \Rightarrow \text{If } r \rightarrow \frac{1}{2}r, \quad F \rightarrow \frac{1}{(1/2)^2} = 4 \times F$$

Answer: 4

Mark: 1

Revision Tips:

- Doppler effect applies to both **sound** and **light** (in astronomy → redshift/blueshift).
- Redshift** helps us measure velocity and distance of distant galaxies using Hubble's Law.
- Newton's Law of Gravitation**: inverse square law – halving distance increases force by 4x.